

Scissor Life Design

by

Tinashe Shoniwa

from Bulawayo, Zimbabwe



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Checked by: Gift Chidewu 32824

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1 Introduction

A scissor lift table is a device that is used to lift or lower heavy loads. Its an essential tool in many industries because of its range of uses. This report will focus on the analysis and calculation of forces and stresses within a scissor lift table under specified load conditions. The study involves determining reaction forces, analyzing member stresses.

2 Objective

The main goals include:

- Determining the reaction forces at the supports and pin joints.
- Calculating the internal forces in all members of the scissor lift.
- Evaluating the stresses in each pin.

3 Calculations

- Table Length: 1000 mm
- Table Width: 500 mm
- Table Thickness: 500 mm
- Length of Leg: 900 mm
- Leg width is 15 mm thick
- Closed height: 350 mm
- Open height: 750 mm
- Applied Force: 5 kN
- Material : Structural Steel, density is 7800 kg/m³

Firstly, we will calculate the angles between the table and the legs at the maximum and minimum height. To do this, we use the Pythagorean theorem, I have attached an image to illustrate.

$$\arctan(\theta) = \frac{350 \text{ mm}}{900 \text{ mm}} = 21.25^\circ$$
$$\arctan(\theta) = \frac{750 \text{ mm}}{900 \text{ mm}} = 39.8^\circ = 40^\circ$$

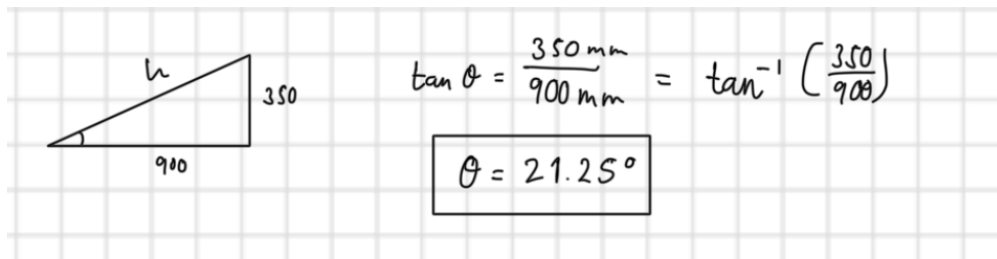


Figure 1: Angle for 350mm height

Since the table will be a shell like structure, we will calculate the volume using,

$$\text{Total Volume} = \text{Outer Volume} - \text{Inner Volume}$$

$$\text{Outer Volume} = 1000 \text{ mm} \times 500 \text{ mm} \times 80 \text{ mm} = 4 \times 10^7 \text{ mm}^3$$

$$\text{Inner Volume} = 990 \text{ mm} \times 490 \text{ mm} \times 60 \text{ mm} = 2.91 \times 10^7 \text{ mm}^3$$

$$\text{Total Volume} = 4 \times 10^7 \text{ mm}^3 - 2.91 \times 10^7 \text{ mm}^3 = 1.08 \times 10^7 \text{ mm}^3$$

With these parameters in place, we now calculate the total mass of the table.

$$m = \rho v \quad (1)$$

where:

- m is the mass of the object in kg ,
- ρ is the density of the object in kg/m^3
- v is the volume in m^3 .

$$m = 7800 kg/m^3 * 1.08 * 10^7 mm^3 = 8.49 kg$$

$$W = mg \quad (2)$$

$$W = 8.49 kg * 9.81 m/s^2 = 83.3 N$$

- m is the mass of the object in kg ,
- g is the gravitational acceleration of the object in m/s^2

3.1 Case 1: Load at 250mm and 350mm height

The load is first applied at 250mm. The diagram shows what the full sketch looks like.

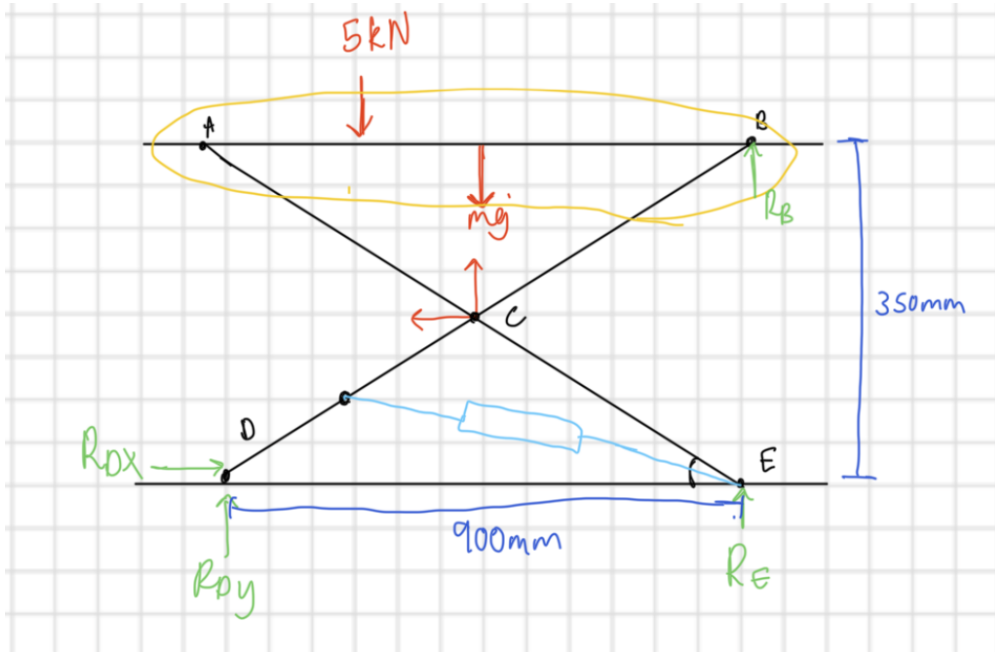


Figure 2: Case 1 Sketch

After 'cutting' the yellow part, we get the following free body diagram AB,

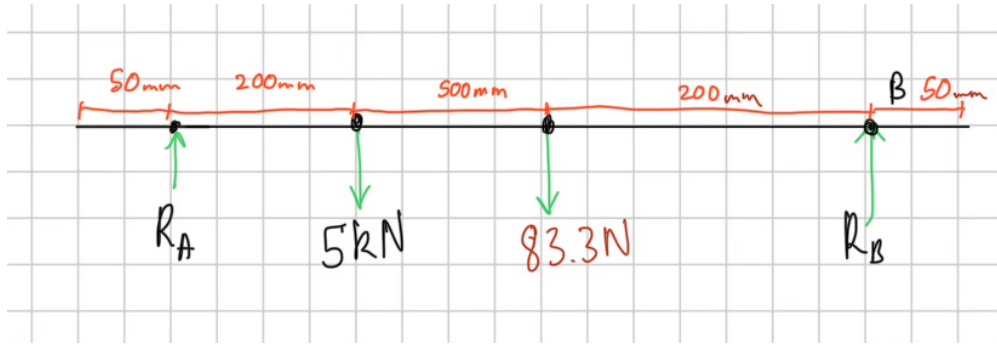


Figure 3: FBD of AB

$$\sum F_x = 0$$

$$\sum F_y = 0 = R_a - 5KN - w + R_b$$

$$R_a + R_b = 5083.4N$$

$$\sum M_A = 0 = -5KN * 200mm - 83.3N * 700mm + R_b * 950mm$$

$$R_b * 950mm = 1058310Nmm$$

$$R_b = \frac{1058310N}{950mm} = 1114N$$

$$R_a = 3969.4N$$

After 'cutting' again, we get FBD ACE,

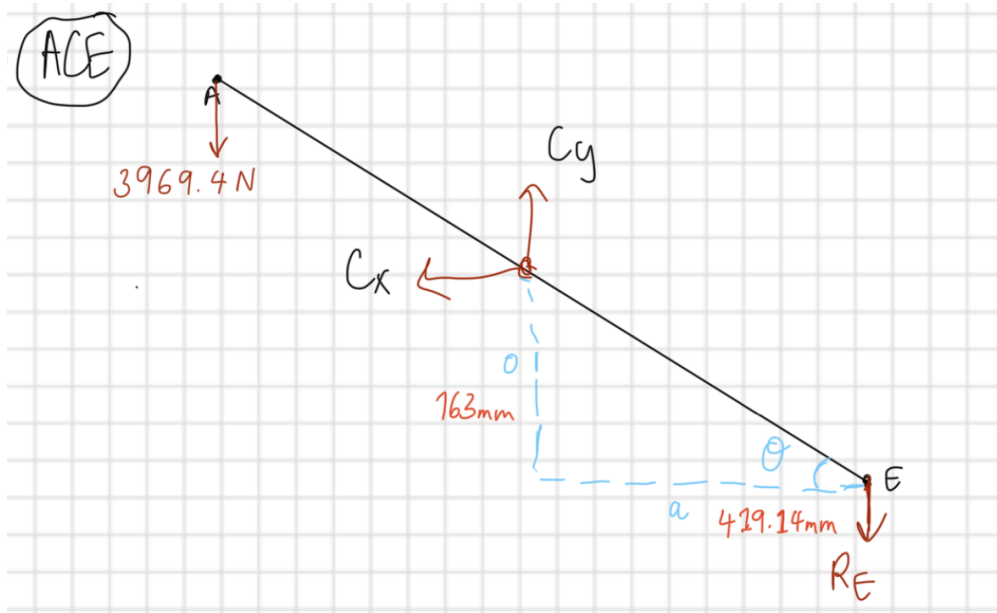


Figure 4: FBD of ACE

$$\sum F_x = 0$$

$$\sum F_y = 0 = C_y + R_E - 3969.4N$$

$$\sum M_A = 0 = 3969.4N * 419.14mm - R_E * 419.14mm$$

$$R_E = 3969.4N$$

$$C_Y = 2 * 3969.4N = 7938.7N$$

After 'cutting' the sketch again, we get the FBD BCD,

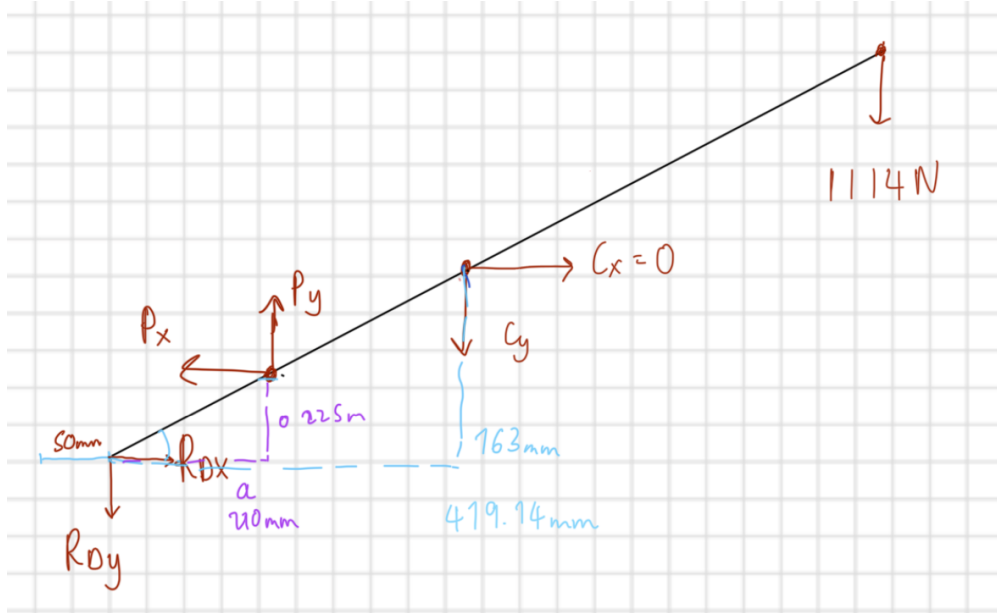


Figure 5: FBD of BCD

$$\sum F_x = R_{Dx} - P_x = 0$$

$$R_{Dx} = -P_x$$

$$\sum F_y = 0 = R_{Dy} + P_y - 7938.7N - 1114N$$

$$R_{Dy} + P_y = 9052.7N$$

$$\sum M_D = 0 = P(\cos(21.25) * 225mm + \sin(21.25) * 210mm) - 7938.7N * 419.14mm - 1114N * 900mm$$

$$P = \frac{7938.7N * 419.14mm + 1114N * 900mm}{\cos(21.25) * 225mm + \sin(21.25) * 210mm} = 15149.8N$$

$$P_x = 15149.8N * \cos(21.25) = 14119.7N$$

$$P_y = 15149.8N * \sin(21.25) = 5490.8N$$

$$C_X = 0$$

$$R_{Dx} = -14119.7N$$

$$R_{Dy} = 3561.8N$$

3.2 Case 2: Load at 250mm and 750mm

First we do the sketch of the whole diagram,

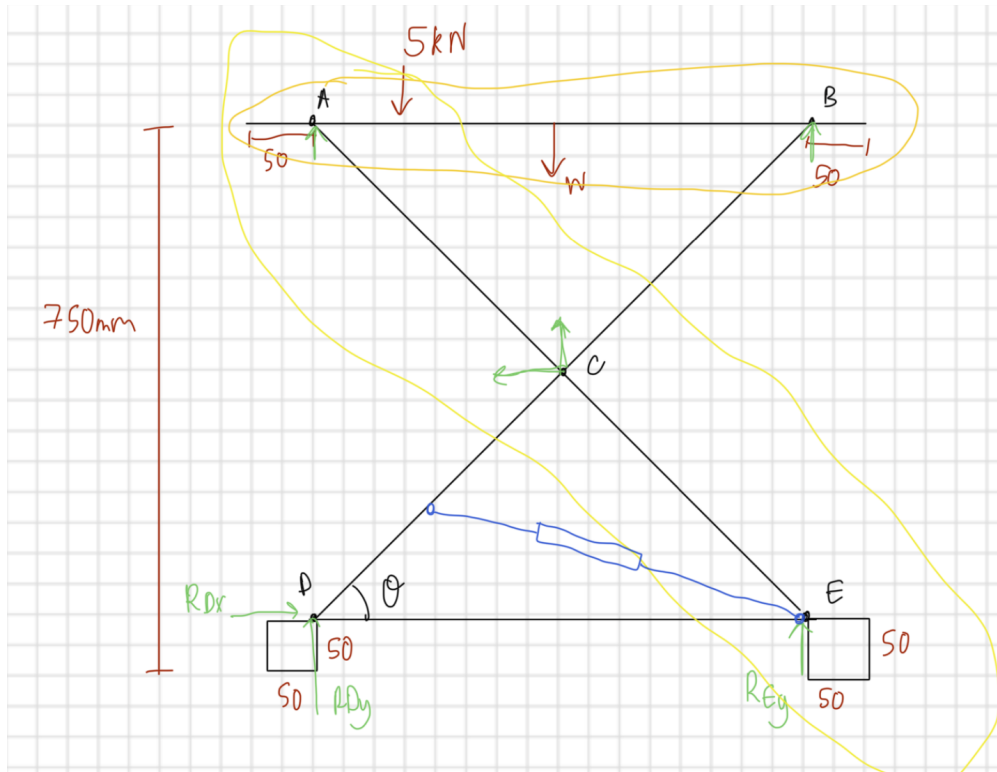


Figure 6: Case 2 Sketch

Then we introduce cuts to AB and get the following figure,

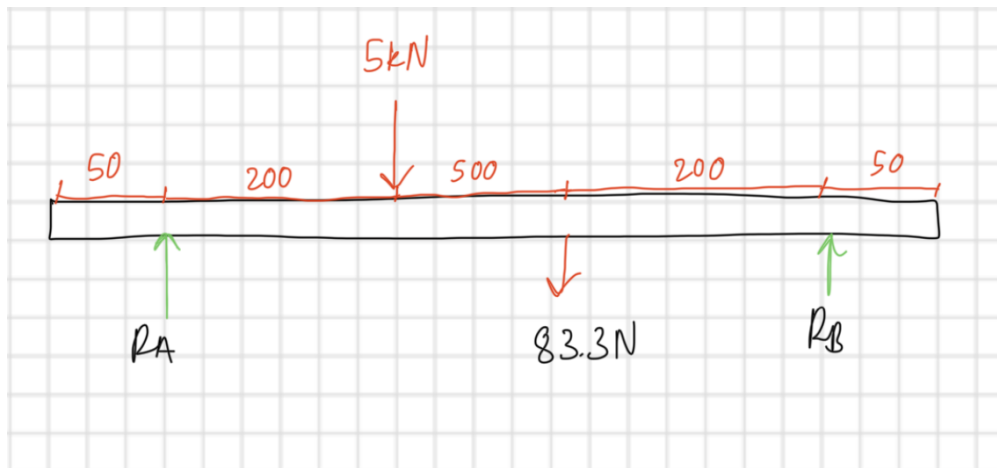


Figure 7: FBD of AB

$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 = R_a - 5KN - w + R_b \\ R_a + R_b &= 5083.4N \\ \sum M_A &= 0 = -5KN * 200mm - 83.3N * 700mm + R_b * 950mm \\ R_b * 950mm &= 1058310Nmm \\ R_b &= \frac{1058310N}{950mm} = 1114N \\ R_a &= 3969.4N\end{aligned}$$

Then we do a cut on one of the legs to get,

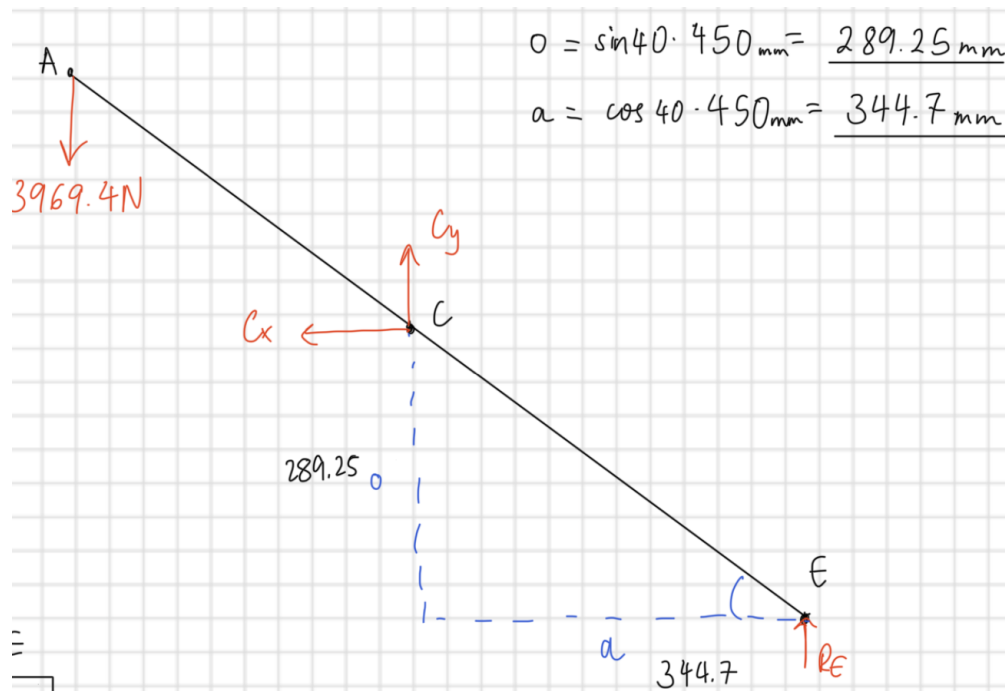


Figure 8: FBD of ACE

$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 = C_y + R_E - 3969.4N \\ \sum M_A &= 0 = 3969.4N * 419.14mm - R_E * 419.14mm \\ R_E &= 3969.4N \\ C_Y &= 2 * 3969.4N = 7938.7N\end{aligned}$$

Then we also do a cut for the other leg to get,

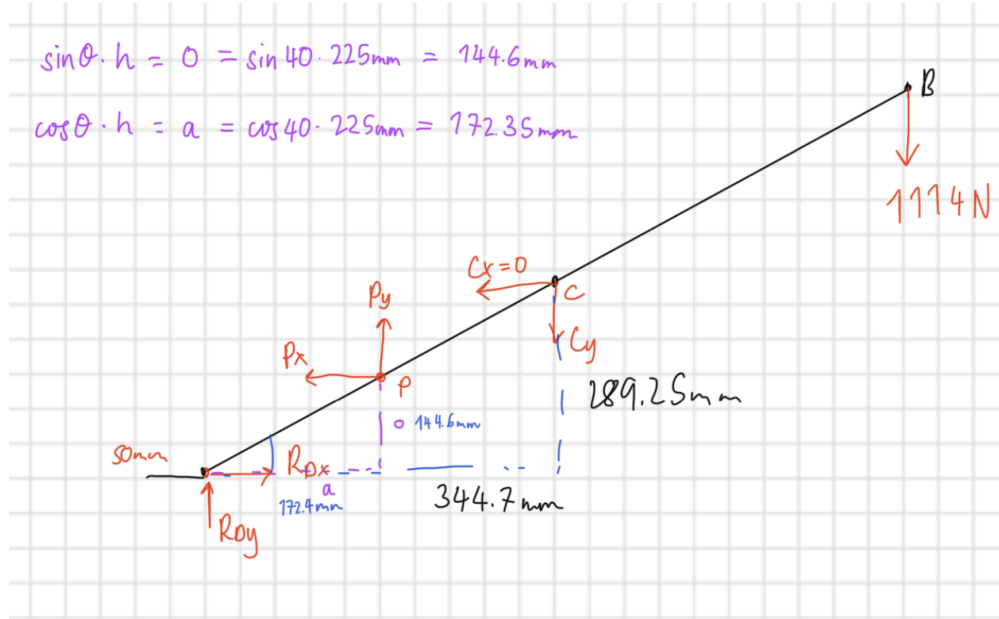


Figure 9: FBD of BCD

$$\sum F_x = R_{Dx} - P_x = 0$$

$$R_{Dx} = -P_x$$

$$\sum F_y = 0 = R_{Dy} + P_y - 7938.7N - 1114N$$

$$R_{Dy} + P_y = 9052.7N$$

$$\sum M_D = 0 = P(\cos(40) * 172.4mm + \sin(40) * 144.6mm) - 7938.7N * 344.7mm - 1114N * 900mm$$

$$P = \frac{7938.7N * 344.7mm + 1114N * 900mm}{\cos(40) * 172.4mm + \sin(40) * 144.6mm} = 16618.6N$$

$$P_x = 16618.6N * \cos(40) = 12730.6N$$

$$P_y = 16618.6N * \sin(40) = 10682.3N$$

$$C_x = 0$$

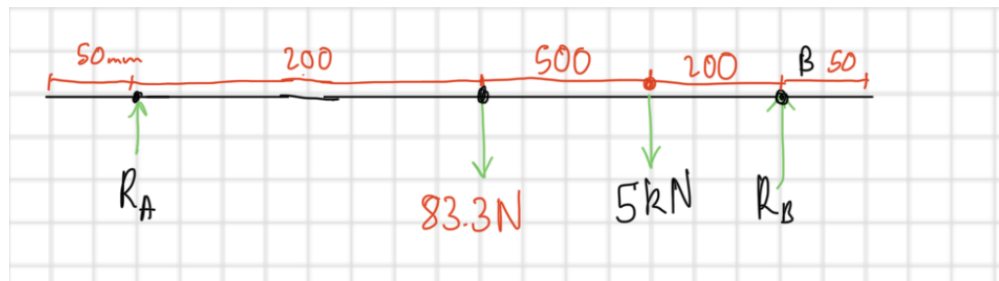
$$R_{Dx} = -12730.6N$$

$$R_{Dy} = -1629.6N$$

We first introduce a sketch of the overall lift,

[illegible]

Then we do another FBD of AB,


$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 = R_a - 5KN - w + R_b \\ R_a + R_b &= 5083.4N \\ \sum M_A &= 0 = -5KN * 750mm - 83.3N * 500mm + R_b * 950mm \\ R_b * 950mm &= 3766660Nmm \\ R_b &= \frac{3766660Nmm}{950mm} = 3964.9N \\ R_a &= 1118.3N\end{aligned}$$

Then we draw FBD of leg ACE,

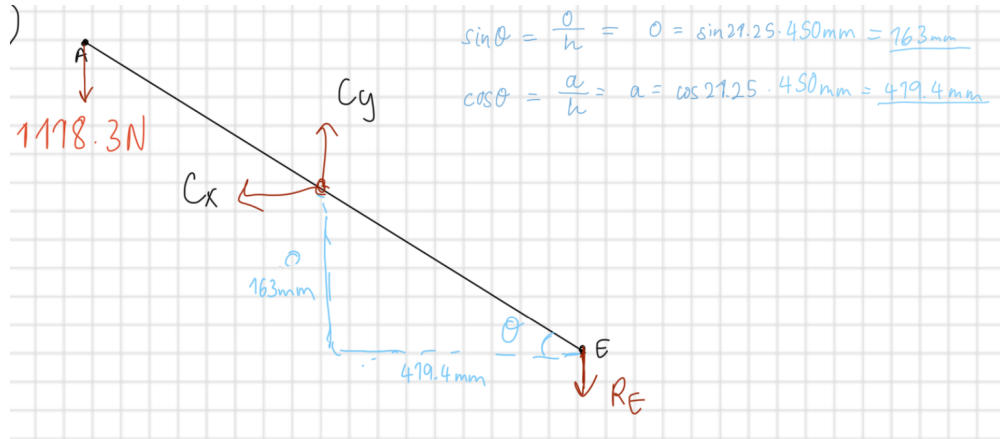


Figure 12: FBD of ACE

$$\sum F_x = 0$$

$$\sum F_y = 0 = C_y + R_E - 1118.3N$$

$$\sum M_A = 0 = 1118.3N \cdot 419.14mm + R_E \cdot 419.14mm$$

$$R_E = 2399.7N$$

$$C_Y = 2 \cdot 2399.7N = 4799.5N$$

Then we also draw another free body diagram of leg BCD,

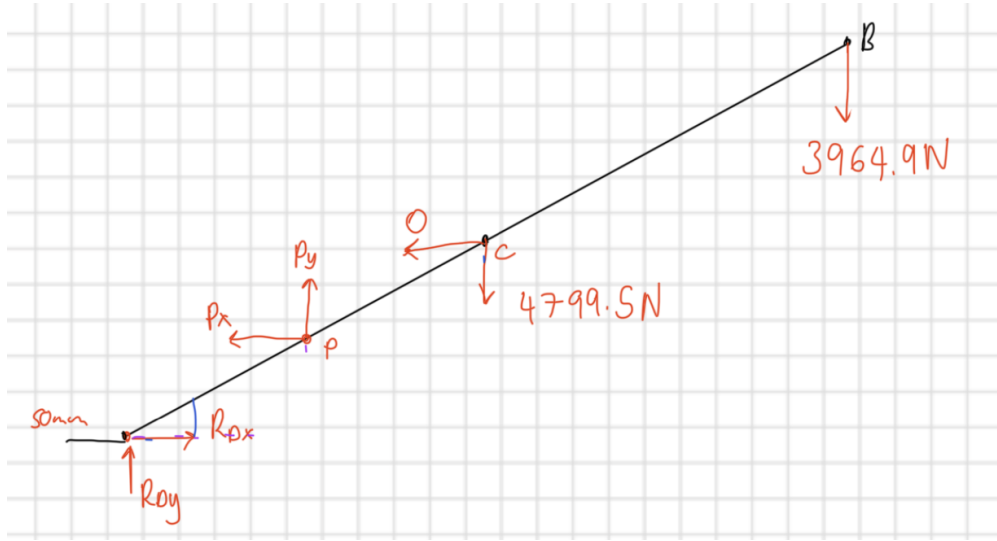


Figure 13: FBD of BCD

$$\sum F_x = R_{Dx} - P_x = 0$$

$$R_{Dx} = -P_x$$

$$\sum F_y = 0 = R_{Dy} + P_y - 4799.5N - 3964.9N$$

$$R_{Dy} + P_y = 8764.4N$$

$$\sum M_D = 0 = P(\cos(21.25) * 144.6mm + \sin(21.25) * 172.4mm) - 4799.5N * 419.14mm - 3964.9N * 900mm$$

$$P = \frac{3964.9N * 900mm + 4799.5N * 419.14mm}{\cos(21.25) * 144.6mm + \sin(21.25) * 172.4mm} = 28288.9N$$

$$P_x = 28288.9N * \cos(21.25) = 26365.5N$$

$$P_y = 28288.9N * \sin(21.25) = 10252.9N$$

$$C_X = 0$$

$$R_{Dx} = -26365.5N$$

$$R_{Dy} = -1488.6N$$

Finally, we draw a sketch of the scenario as follows,



A horizontal beam is shown with the following dimensions and forces:

- Dimensions (from left to right): 50 mm, 450 mm, 250 mm, 200 mm, 50 mm.
- Forces:
 - Reaction force R_A acting upwards at the left end.
 - Reaction force R_B acting upwards at the right end.
 - A downward force of 5 kN acting at the right end.
 - A downward force of 83.3 N acting at the 450 mm mark from the left.

Figure 15: FBD of AB

$$\begin{aligned}
\sum F_x &= 0 \\
\sum F_y &= 0 = R_a - 5KN - w + R_b \\
R_a + R_b &= 5083.4N \\
\sum M_A &= 0 = -5KN * 750mm - 83.3N * 500mm + R_b * 950mm \\
R_b * 950mm &= 3791650Nmm \\
R_b &= \frac{3791650Nmm}{950mm} = 3991.2N \\
R_a &= 1092.1N
\end{aligned}$$

Then the FBD of the leg ACE is shown as follows,

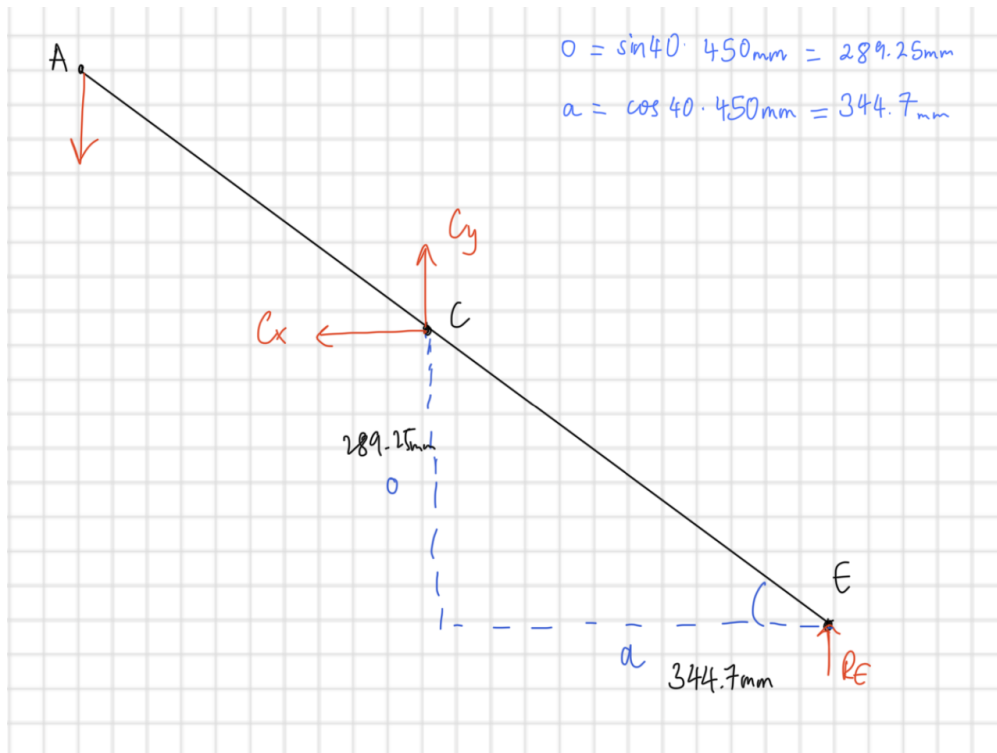


Figure 16: FBD of ACE

$$\begin{aligned}
\sum F_x &= 0 \\
\sum F_y &= 0 = C_y + R_E - 1092.1N \\
\sum M_C &= 0 = 1092.1N * 344.7mm + R_E * 344.7mm \\
R_E &= 1092.1N \\
C_Y &= 2 * 1092.1N = 2184.2N
\end{aligned}$$

Then we conclude with the FBD of BCD,

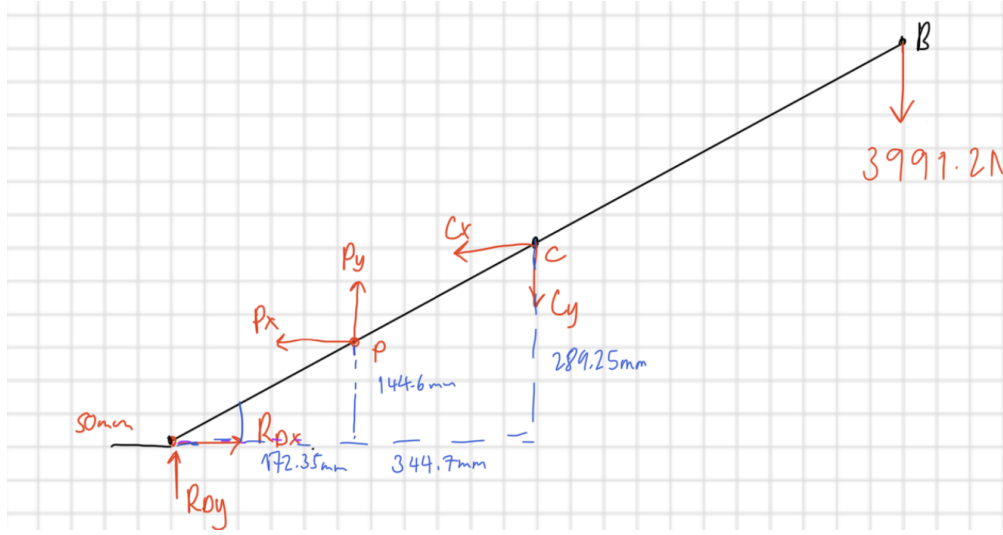


Figure 17: FBD of BCD

$$\sum F_x = R_{Dx} - P_x = 0$$

$$R_{Dx} = P_x$$

$$\sum F_y = 0 = R_{Dy} + P_y - 7938.7N - 1114N$$

$$R_{Dy} + P_y = 9052.7N$$

$$\sum M_D = 0 = P(\cos(40) * 172.4mm + \sin(40) * 144.6mm) - 2184.2N * 344.7mm - 3991.2N * 900mm$$

$$P = \frac{2184.2N * 344.7mm + 3991.2N * 900mm}{\cos(40) * 172.4mm + \sin(40) * 144.6mm} = 19309.8N$$

$$P_x = 19309.8N * \cos(40) = 14792.2N$$

$$P_y = 19309.8N * \sin(40) = 12412.1N$$

$$C_x = 0$$

$$R_{Dx} = 14792.2N$$

$$R_{Dy} = -9052.7N$$

3.5 Assumptions

To make the calculations simpler, i neglected friction and i also assumed the piston as a stroke for easier calculation

4 Stress Analysis

- Leg connector pin Below I have attached a sketch profile of the leg connector that connects the legs to the table. All values are in mm.

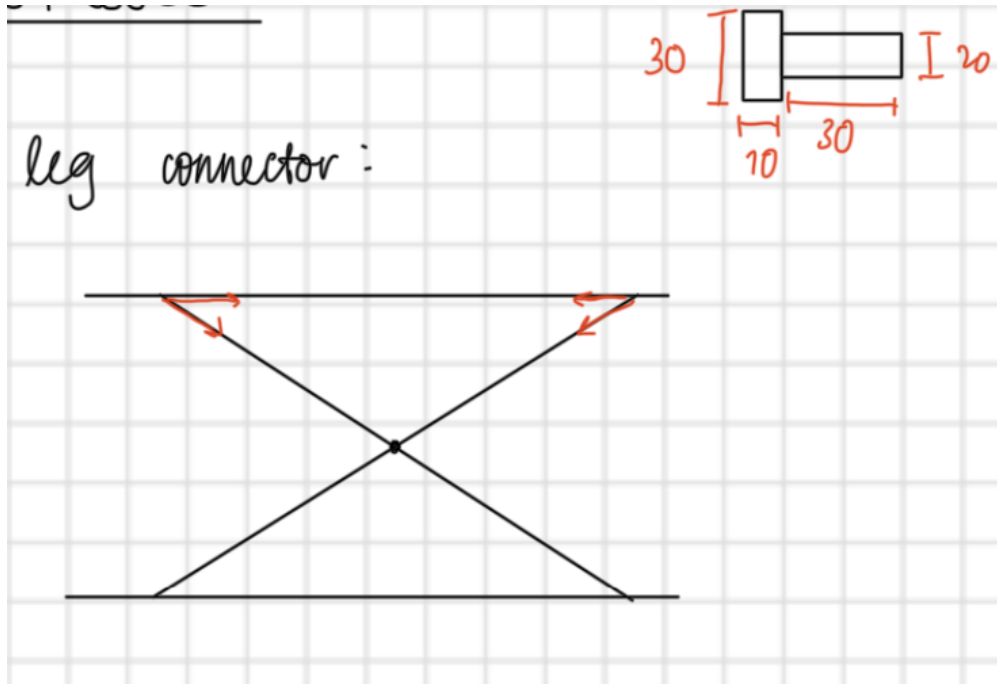


Figure 18: Sketch Profile of Pin

4.1 Stresses at max height(350mm)

To calculate shear stress using the formula;

$$\tau = \frac{F}{A} \quad (3)$$

where;

- F is the applied force in N
- A is the area in mm^2

To find the area, using the radius of 10mm

$$Area = \pi r^2 = 314.16mm^2 \quad (4)$$

Then using the applied force of

$$\frac{5083.3N}{2} = 2541.5N$$

This gives us

$$\tau = \frac{2541.5N}{314.16mm^2} = 8.09MPa$$

4.2 Stresses in the legs

We then use the flexure formula:

$$\sigma = \frac{M * c}{I} \quad (5)$$

where

- M is the maximum bending moment
- c is the distance from neutral axis
- I is the moment of inertia

$$\frac{19309.8}{\frac{1}{12} * 15 * 2 * 114} = 67.75 Nmm^2$$

The shear stress in each leg is as follows;

$$\tau = \frac{7012.6N}{1359} = 5.19 MPa$$

4.3 Weld Calculations

To calculate the weld strength, we will use the formula

$$F = \tau * t * L \quad (6)$$

where

- τ , is the allowable stress
- t is the throat thickness
- L is the length of the weld

I have decided not to include them into this document as I only intended to do the Solidworks Design for practice and I am not planning on doing the exam this semester.

5 References

Olenin, G. (2016). Degree Programme in Mechanical Engineering and Production Technology Design of hydraulic scissors lifting platform. [online] Available at: https://www.theseus.fi/bitstream/handle/10024/113124/Olenin_Georgy.pdf.